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Combating COVID-19 with charisma: Evidence on governor speeches in the United States

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ABSTRACT

Using field and laboratory data, we show that leader charisma can affect COVID-related mitigating behaviors. We coded a panel of U.S. governor speeches for charisma signaling using a deep neural network algorithm. The model explains variation in stay-at-home behavior of citizens based on their smart phone data movements, showing a robust effect of charisma signaling: stay-at-home behavior increased irrespective of state-level citizen political ideology or governor party allegiance. Republican governors with a particularly high charisma signaling score impacted the outcome more relative to Democratic governors in comparable conditions. Our results also suggest that one standard deviation higher charisma signaling in governor speeches could potentially have saved 5,350 lives during the study period (02/28/2020–05/14/2020). Next, in an incentivized laboratory experiment we found that politically conservative individuals are particularly prone to believe that their co-citizens will follow governor appeals to distance or stay at home when exposed to a speech that is high in charisma; these beliefs in turn drive their preference to engage in those behaviors. These results suggest that political leaders should consider additional "soft-power" levers like charisma—which can be learned—to complement policy interventions for pandemics or other public heath crises, especially with certain populations who may need a "nudge."

Introduction

Governor: *"steersman, pilot"* (origin: Ancient Greek: κυβερνήτης; kubernḗtēs)

In times of turmoil or grave threat, a leader—whether a skipper of a boat, a pilot of an airplane, or a head of a state—is propelled to the fore. The importance of leadership among US governors and the federal government in facing the COVID-19 pandemic has been widely conveyed in the media. From CNN and Forbes to the Guardian and the Washington Post, news outlets have stressed why political leaders' communication is key for successfully managing the pandemic.[1] Whereas journalistic accounts have some value and insights, they draw on anecdotal evidence. However, from a policy point of view, can we scientifically establish if leaders' qualities really matter, especially at the governor level, where discretion in policy implementation is large?

This question could not be of greater significance and urgency, given the pandemic's colossal human, economic and social costs (see Bonardi et al., 2020) and the fact that stay-at-home orders do not seem to be, overall, as effective as hard measures like shutting down business (e.g., Brauner et al., 2020). If more effective governor

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[1] See e.g. https://edition.cnn.com/2020/04/05/politics/governors-national-spotlight/index.html; https://www.forbes.com/sites/carminegallo/2020/03/19/how-new-york-governor-andrew-cuomo-balances-calm-with-the-need-for-drastic-measures-in-covid-19-updates/?sh=4549abd95bcc; https://www.theguardian.com/world/2020/mar/23/cuomo-wins-praise-for-wisdom-amid-coronavirus-crisis-as-trump-blusters; https://www.washingtonpost.com/lifestyle/style/andrew-cuomo-during-the-covid-19-crisis-is-the-same-as-ever-with-one-big-difference-people-like-him/2020/03/28/11a89a0a-6fd7-11ea-b148-e4ce3fbd85b5_story.html.

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communication could—even to a small extent—impact citizens' choice to stay at home or undertake other measures to mitigate the effect of COVID-19 (e.g., physically distance) and comply with best-practice health measures, the societal gain of governor speeches would be enormous. Effective communication can be thought of as a non-pharmaceutical intervention (NPI) that helps to combat the spread of the virus. Such gains from leader communication are especially salient early on in large-scale crises like a pandemic where pharmacological interventions (e.g., vaccines) are not yet available.

The above question is also of general importance beyond the current COVID-19 pandemic: Knowing whether communication matters in a large-scale viral, economic, and societal crisis will provide crucial policy lessons for future crises (e.g., see Tortola & Pansardi, 2019). An important aspect of the leader's qualities is their ability to signal information clearly in a way that can reduce information asymmetries, help solve coordination problems, reduce selfish actions, and preserve the public good (Bastardoz & van Vugt, 2019). A "soft" and non-legal means of influence that has strong effects especially in ambiguous situations, and which should matter in the current milieu, is leader charisma (Antonakis, 2021).

In this article, we offer an empirical contribution by investigating whether US governor charisma, operationalized via charismatic verbal signaling in speeches can help combat COVID-19. In the first study, we draw on a deep learning algorithm to code charisma signaling content of 350 US governor speeches during the pandemic and estimate how a measure of governor charisma explains variation in stay-at-home behavior; a crucial element of physical distancing guidelines and strategies early on in the pandemic. To help unpack the state-level results in more depth, the second part of the article is dedicated to a lab experiment studying the causal impact of treatments in which we manipulate governor charisma and observe participants' incentivized choices related to beliefs about distancing and stay-at-home behavior as well as their own willingness to distance. Note, by "incentivized choices" we mean the fact that the participant's payout is greater when providing more accurate estimates of others' overall behavior.

In our contribution, we provide a strong test of whether charisma signaling matters, particularly in an ecologically valid field context with real world outcomes. The field context provides us with an unusual level of control and standardization (Bamberger & Pratt, 2010). Focusing on governors we can partial out many confounds at the state level, account for how the pandemic unfolds with respect to time and provide a common platform on which we observe the transmission of charisma signals to citizens (via governor briefings) across 50 comparable contexts (i.e., states). Importantly, we are also able to observe the actual geographical mobility of citizens unobtrusively and objectively. Showing that charisma signals in governor communication may significantly slow down geographical mobility and may thereby prevent a substantial number of deaths from COVID-19 has important implications for policy. We also conduct an incentivized lab experiment to examine the extent to which exposure to appeals for distancing alters the incentivized beliefs and participants' willingness to distance as a function of charisma signaling. Because it appears clear in this age that large cleavages exist between the preferences of politically liberal and politically conservative individuals with respect to pandemic policy levers (Gollwitzer et al. 2020), we also examine whether citizens' political ideology (i.e., beliefs about the proper order of society and how it can be achieved, Erikson and Tedin 2003, p. 64) matters for the effect of charismatic appeals. Finding non-legal ways—like persuasive communication that go beyond enforceable legal injunctions—to influence citizens is thus a policy imperative.

Why charisma should matter

Charisma can be defined as "value-based, symbolic, and emotional-laden signaling" (Antonakis et al., 2016, p. 304). Note, "charisma" refers to the charismatic effect occurring on the level of the observer. Throughout the article so far, however, and for the purpose of this study, we refer to "charisma signaling" to make a distinction between (a) the charismatic effect on the level of the observer, which occurs from exposure, in part to the charisma signals and a willingness of the observer to follow, and (b) the signals per se, which may or may not trigger attributions of charisma (depending on the values of the leader, the context, etc. – for details on this distinction, see Antonakis et al., 2016 and Bastardoz, 2020). Thus, when referring to variation in charismatic signaling content of the speech for the context of our article we refer to "charismatic signals" or a "charisma score." Importantly, speeches with higher charisma signaling content, however, are generally judged to be more charismatic by observers (e.g., Antonakis et al., 2021; 2011; Meslec et al., 2020), as the results of our Study 2 show as well.

Charisma is posited to help engender commitment to a message, arouse emotions among followers, and stimulate actions that benefit collectives (Shamir et al., 1993). To garner this social influence, charismatic leaders use communication techniques to create a vivid image of their message, to highlight the saliency of the mission as well as its moral imperative, and to increase the psychological identification with followers; in this way charismatic leaders can affect individual commitment and motivation as well as help increase coordination of follower actions (Antonakis et al., 2016). More specifically, research has distilled a set of verbal and non-verbal communication techniques, also known as "charismatic leadership tactics," that leaders can use to appear more charismatic (Den Hartog & Verburg, 1997; Frese et al., 2003) and increase individuals' identification with the leader message. Importantly, these techniques encapsulate communication behaviors that can be measured objectively without the typical observer bias that plagues questionnaire measures (Banks, Woznyj & Mansfield, 2021); measures which solicit variation in observer perceptions and can be affected by a host of omitted variables. Using perceptual measures, if modeled as the independent variable, cannot ensure consistent estimation of model parameters because of endogeneity bias (Fischer & Sitkin, 2023; Fischer, Hambrick, Sajons, & Van Quaquebeke, 2020). Importantly, charisma can be manipulated in lab and field settings and its economic effect is equivalent to that of high-powered bonuses (e.g., Antonakis et al., 2021; Meslec et al., 2020).

Whereas early work stressed charisma as an innate quality of "larger-than-life" leaders (Weber, 1947), contemporary research focuses on charismatic communication behaviors, which is also a learnable skill. Indeed, using random assignment to a training intervention in a Swiss context, Antonakis, Fenley, and Liechti (2011) found that individuals in the experimental group were rated as more charismatic, more competent, and as possessing more prototypical leader qualities three months later by their subordinates and coworkers (see also Frese et al., 2003). These findings are critical not only because they show that charisma is not some mystical quality, but because they illuminate the anatomy of the concept of charisma and offer guidance on how to manipulate it, measure it, and master it.

Specifically, leaders can enact charismatic communication behaviors by framing the message in a vivid manner, providing substantive moral arguments to identify strategic imperatives, and mirroring collective sentiments to promote identification with the leader (Jacquart & Antonakis, 2015). Charismatic leaders do so by harnessing the power of various rhetorical techniques. For instance, leaders can create a visual using metaphors, stories, and anecdotes. These features trigger imagery by connecting symbolic meaning or relatable anecdotes to followers' realities. Leaders can also create an intrigue; a puzzle to be solved by the listener through rhetorical questions. Contrasts help draw the audience in by creating a dramatic effect and show a course of action by pitting right against wrong, or good against bad. Finally, leaders can frame the message through repetitions or lists to portray a pattern and sense of completeness in the message (Antonakis et al., 2011; Jacquart & Antonakis, 2015).

Leaders can also give substance to their core message by appealing to moral conviction, and to the sentiment of the collective (e.g., their hopes, fears, and aspirations). Such appeals infuse values and virtues into the message, emphasizing what is right to do, the moral responsibility of the individual, and the collective identity of the group. Finally, leaders can also direct followers' action by setting ambitious goals and instill hope and confidence that these goals can be achieved (see Antonakis et al., 2016).

Although nonverbal aspects are important too, they play a much lesser role in inducing the charismatic effect, even no role when the verbal aspects are controlled (e.g., Tur et al., 2021, Table 3). If governors can change their citizens' behavior simply through their use of charisma, then boosting governors' charisma—through training and development of these specific techniques or using speech writers to help them (cf. Jacquart & Antonakis, 2015)—represents a powerful lever for combating COVID-19, or in other public health threats. In line with these ideas, recent experimental studies show that charismatic leaders can stimulate individual performance—even to an extent that matches the effects of economic incentives (Antonakis et al., 2021; Meslec et al., 2020)—see also Fest, Kvaløy, Nieken, & Schöttner (2021). A recent study by Boulu-Reshef and colleagues (2020) adds more credence to this position, demonstrating that leadership communication can alleviate free-rider problems.

Together these results from existing studies complement our expectations outlined here in supporting that the "soft power" leaders hold in their strategic and balanced use of charismatic communication can help coordinate collective action and safeguard the public good—in the case of COVID-19: public health. On this basis, we test:

Hypothesis 1. Governor charisma signaling increases compliance with public health guidelines through increased stay-at-home behavior.

Whereas Hypothesis 1 captures the expectation of a main effect of charisma signaling on stay-at-home behavior, it does not account for potential complexities at the individual level. In fact, the population of each US state is potentially quite heterogeneous in their reactivity to charismatic communication, especially along political ideological divides. Political ideology can be defined as a "set of beliefs about the proper order of society and how it can be achieved" (Erikson & Tedin, 2003, p. 64), with the superstructure "liberal–conservative" often used to demark fault lines on social and political issues in the United States (Jost, Federico, & Napier, 2009). Accounting for this distinction is important because research during the COVID-19 pandemic has consistently shown greater support for public health measures among politically liberal individuals (e.g., Gollwitzer et al., 2020) and greater reluctance among politically conservative individuals.

Thus, people may react to governor requests not only as a function of the governor's level of charisma signaling but also by their individual preferences or demographics, as well as their beliefs about what others may do. The latter point is particularly important because some individuals (e.g., politically conservatives) may not be willing to incur a cost related to their own behavior if they know others may not incur the cost as well; in line with the literature on "conditional cooperation" (e.g., see Frey & Meier, 2004). Of course, there are other intricacies involved with respect to how COVID-19 is viewed as a function of political ideology, particularly with respect to science denialism, and how personal liberties may conflict with what is best for the public good, as well as public health policy (e.g., Baccini & Brodeur, 2021; Calvillo, Roos, Garcia, Smelter, & Rutchick, 2020; Hamilton & Safford, 2021; Samore, Fessler, Sparks, & Holbrook, 2021; Wang, Devine, & Molina-Sieiro, 2021).

Furthermore, individuals who are better informed on the evolution of the pandemic—or more apt to adopt scientific guidelines—may, for example, require less convincing for adopting preventive public health measures such as distancing or staying at home (cf. Bayram & Shields, 2021; Calvillo et al., 2020). For such a subgroup of the population, charisma signaling may not make much of a difference because this group may adopt COVID-mitigating guidelines regardless, even after non-charismatic appeals. In contrast, for COVID-19-sceptics—often immune to "sterile" science talk and hard facts—charismatic communication may help a great deal for convincing them to stay at home or engage in other costly behaviors; perhaps the message only passes if communicated by someone whose ideology overlaps with the receiver of the message (cf. Koetke, Schumann, & Porter, 2021). To explore the potential heterogeneous response along individuals' political ideology, and in absence of theory to guide us on how charisma signaling affects preferences as a function of ideology in the COVID-context, we also investigate the following research question:

Research question 1: Is the effect of governor charisma signaling on compliance with public health guidelines through increased willingness to stay at home moderated by political ideology?

We will examine several issues here, including: does charisma signaling have a stronger or weaker effect for individuals holding conservative ideological beliefs? Does charisma signaling work better as a function of governor political allegiance? Is sender-receiver political ideology congruence required for the charismatic message to be acted on?

Overview of studies

We conducted two studies to test our hypothesis and to explore our research question. In Study 1, we test the impact of governor charisma signaling on citizens' stay-at-home behavior using 350 coded COVID-19 press briefing speeches by all 50 US governors combined with population mobility patterns from anonymized smartphone data during the early part of the pandemic. Because Study 1 yields an aggregate analysis at a collective level, it is impossible to zoom in on individual choices to explore the role of population heterogeneity in US states and the potential sensitivity of individuals to charismatic communication. Moreover, even though we account for governor fixed effects, it may be possible that omitted variables in our specification bias the estimates. Also, even if the estimated effect is accurate, the fact that the field data is anonymized at the state-level may hide substantial heterogeneity in citizens' responses.

For these reasons, we take a more fine-grained approach in Study 2 using a laboratory experiment among 661 US adults in which we manipulate governor charisma signaling in an incentivized vignette experiment to control and investigate parameters of decision-making related to beliefs about and willingness to stay at home. This study offers evidence on the effect of charisma on two variables; (a) y_1 – participants' beliefs about how others would react to the speech; here, we incentivized participants to be accurate, and (b) y_2 – participants' own willingness to engage in stay-at-home behaviors following exposure to the speech.

All study materials, including data, code, and questionnaires, as well as transparency reports (Aczel et al., 2020) can be accessed here: https://osf.io/mj2vs. Study 2 protocol and design were preregistered with the Open Science Framework prior to data collection and can be accessed here: https://osf.io/ghejp and here: https://osf.io/hbpmr. For Study 2, a main effect of charisma signaling on willingness to stay at home was preregistered as a confirmatory analysis, wherein we declared our intention to explore the potential moderating role of political ideology.

Study 1

Method

Procedure

For Study 1, we sampled seven COVID-19 press briefing speeches for each of the 50 United States governors ($n = 350$) to measure charisma signaling in their public health messages. Given the large cluster

size ($G = 50$) and having balanced panel data ($n = 7$) our sample size is sufficiently large for consistent estimation and inference (Cameron & Miller, 2015; see also Antonakis et al., 2021). Our sample size has adequate power to detect a medium effect (Scherbaum & Ferreter, 2009), and is practically feasible given the extensive efforts needed to identify, transcribe, and code speech recordings. Monte Carlo simulations (cf. Appendix A) similarly support the notion that our sample size is sufficient and that our estimator remains reliable; even when we simulate more extreme data specificities.

The COVID-19 press briefing speeches cover a timeframe of approximately 2 months, March through April 2020, and were selected at random within weeks. This time period encompasses the crucial first months of the outbreak of SARS-CoV-2 in the United States, including the timeframe during which most states introduced some version of shelter-in-place executive orders as part of promoting compliance with public health guidelines. We combine the data on US governors' speeches with publicly available information on governor and state characteristics, as well as population mobility data from smartphones to estimate changes in stay-at-home behavior over time as a function of governor charisma signaling.

Measures

Charisma signalling

For each of the 350 speeches, we identified a high-quality video recording and trimmed the file to only include the governor's speech and concluding remarks. Speeches were then transcribed by humans and coded sentence by sentence for charismatic signaling using a deep learning machine algorithm (see also Marshall et al., 2022 for a coding approach that uses another theoretical base and machine coding method). This measure yields the total number of charismatic signals in each speech. Prior validation efforts showed that this algorithm performs very well in predicting charismatic signaling coded by humans on a large number of speeches. Appendix B discusses these validation efforts in more detail and provides a technical note on the algorithm. This algorithm has also performed well in measuring charisma signaling in Tweets (see Study 2 in Tur et al., 2021). We also used it in our own Study 2 (see section on "Manipulation Check" and Table 2), where we offer additional evidence to its validity. In addition to the charisma scores, we included the total number of sentences to account for speech length. The algorithm estimates the probability that each of the nine charismatic leadership signals is present in a sentence. Thus, the upper bound (maximum) score a speech can obtain is the number of sentences times nine. The maximum score would be close to impossible to obtain given the difficulty in exhibiting nine charisma signals in each sentence.

Given that speech length (number of sentences) mechanistically determines the upper bound of our charisma measure, we naturally observe a high correlation between the two (cf. Table C1a). However, as we detail in a series of Monte Carlo simulations (cf. Appendix A), our estimator is very stable and unaffected when we control for speech length (see also Appendix D). The simulations also confirm that our estimator reliably detects an effect of charisma signaling given our sample specificities. Finally, the notion that it is the stylistic features indicative of charisma (i.e., the charismatic leadership tactics) that drives the effect is supported by existing experimental work in which researchers manipulate the number of charismatic leadership tactics but hold constant speech length (e.g., Antonakis et al., 2021; Fest et al., 2021; Meslec et al., 2020).

We also had two humans code the speeches for whether the governors asked residents to stay at home or not (or to mitigate the effect of COVID in other ways). Coders were first trained on two batches of 50 speeches to ensure consistency and we estimated their reliability (Landis & Koch, 1977). On the first batch, independent pre-agreement was 72.00% (expected chance agreement = 51.04%), $kappa = .43$, $SE = .12$, $z = 3.47$, $p = .00052$. Although initial agreement was good, the coders reconciled their ratings until reaching full agreement with the last author intervening for tie breakers. Using updated coding rules, coders independently coded the second batch of speeches reaching substantial agreement of 84.00% (expected chance agreement = 55.76%), $kappa = .64$, $SE = .14$, $z = 4.69$, $p < .0001$; again, remaining discrepancies were reconciled. Each coder then coded the remaining speeches. In 265 (75.71%) cases, the governors explicitly asked residents to undertake COVID-19 mitigating measures, suggesting that our sample represents a highly salient outlet for governors to promote public health preventative behaviors in their state, and that the speeches have as an obvious overall goal to get citizens to keep distance from others in a general sense, and in part by staying at home. Note, whether we include the dummy variable indicating whether the governor makes COVID-19 mitigating appeals in the regression model or not does not change the results of our main models probably because citizens were rather focused on the pandemic at the time in a general sense, and that most governor speeches explicitly asked citizens to change their behavior. Thus, theoretically, whether citizens chose to mitigate the effects of COVID-19 should depend on how persuasive the governor is.

Stay-at-home behaviour

Stay-at-home behavior is measured using a social mobility dataset made freely available for COVID-research by SafeGraph (see Appendix E). We sourced data from the so-called "shelter-in-place dashboard," which captures the percentage of people staying at home all day on a daily basis for all US states. The dashboard is based on anonymized population movement of 45 million smartphone devices and is representative of the US population. To distinguish movement from a reference point, individuals' "home" location is defined as the most common nighttime location to a precision of approximately 100 square meters. This measure does not discriminate between how far or for how long one leaves home but classifies all events of leaving home similarly. Hence, if someone moves outside their common nighttime location for even a short time (e.g., to go to the grocery store half a mile away) during a 24-hour window, they would not be classified as "staying at home." As an example, if our dataset contains 10,000 people residing in Delaware, and 6,000 people moved outside their common nighttime location for any period during a given day—for example, April 1, 2020, the measure will yield a 40% stay-at-home percentage for that day.

We create two measures of stay-at-home behavior based on the date of each speech, both introducing a time lag to ensure correct temporal sequencing with respect to the unraveling of the governor's influence on citizens. To assess almost immediate impact, we first create a 1-day lag to capture the percentage of people staying at home in a given state on the day following a governor's speech. Second, we expect citizens to watch governor speeches in TV or through online media as live stream but also as recorded versions. To allow the effect of the governor's speech to diffuse, we create a measure of the average stay-at-home percentage for the first week following each speech by averaging the percentage of people staying at home all day in a given state over the seven days following a governor's speech. Despite these two variables being highly correlated, their relationship is not fully linear (i.e., it is monotonic positive with a slightly decreasing slope). Given that the two measures are not redundant in that they capture behavior in a different temporal dimension, and that we declared this data at initial submission to the journal, we decided to report results on both measures.

Control variables

We draw on a series of publicly available datasets and information to measure governor and state controls. For governors, we used public bios on governors' websites and Wikipedia to code for gender, age, education, tenure, and party. Issuance and effective date of state shelter-in-place orders were sourced from news articles. We use this information to create an indicator for whether a state adopted a

shelter-in-place (SIP) order or not, as well as a time-varying continuous variable denoting the number of days an order had been in place at the time of a given speech.

To capture the severity of the COVID-19 crisis over time in each state, we rely on case numbers and deaths from the "COVID-19 Data Repository" by the Center for Systems Science and Engineering (CSSE) at Johns Hopkins University (Dong et al., 2020). We aggregated data to the state level and matched case numbers and deaths based on the date of each speech. We control for other time fixed effects as well in two ways: First, we include the number of days that have elapsed since the first governor speech was given (i.e., 2/28/2020). This variable controls for learning effects on the part of the governor (i.e., with time, governors may get better at giving speeches) and other trends (e.g., the population becomes more vigilant). Second, we control for month fixed-effects to capture seasonality.

We also collected data on a host of state characteristics, including percentage of obese people from the Centers for Disease Control and Prevention, GDP from the Bureau of Economic Analysis, population, population density, percentage blacks, Hispanics, people 65 years or older, people under the poverty line, all from the 2019 U.S. Census' American Community Survey (cf. Appendix E for source references). Finally, we collected information on the number of natural disasters in a given state over the past decade from FEMA's "Disaster Declarations Summary" to capture state-level preparedness for disasters. Of note is that these state level regressors are interesting to look at from a basic research point of view; however, they will not alter any inferences we make with respect to the "within" level (i.e., whether a governor speech may affect stay-at-home behavior) given our modeling procedure using fixed-effects, or correlated random effects, estimation.

To examine whether political ideology moderates the charisma signaling effect, we include two measures: (a) the state-level Republican vote share in the 2020 presidential election, and (b) the proportion of individuals who identify as politically conservative or politically liberal, using Gallup tracking data for 2018. The two variables are very highly correlated at polar opposites, $r(50) = -.9041$ (see also Appendix F). Thus, for parsimony and because we estimate a series of interactions (4-way), we created a new variable "conservative," by subtracting the liberal score from the conservative score and ensured that the assumptions of taking difference scores were satisfied (Edwards & Parry, 1993). We report on the justification for using a difference score below.

A list with links to all publicly available datasets used for this study can be found in Appendix E. Table G1 in Appendix G provides an overview of all study variables for the panel regressions. Tables C1a and C1b in Appendix C report report the correlation matrices for study variables at the governor and speech levels.

Estimation

We first use Ordinary Least Squares (OLS) regression analysis with standard errors clustered at the state (governor) level, resulting in a total of 50 clusters. We estimate the following model specification, with the key coefficient of interest being γ_1:

$$y_{i+t,j} = \gamma_1 Charisma_Speech_{i,j} + \varepsilon_{i+t,j} \quad (1)$$

For the dependent variable, we have y, $Stay-at-home\ behavior_{i+t,j}$, which varies at the level of a given state (and governor) j and a given period $i+t$, with i being a given speech, and t being the number of days after the speech (i.e., 1 or 7 days). The main explanatory variable $Charisma_Speech_{i,j}$ varies at the level of speech i and governor j. The specification includes an error term $\varepsilon_{i+t,j}$ which varies at the level of period $i+t$ and state j. We gradually build up the model from OLS (columns 1 and 6 in Table 1a) to use the fixed-effects estimator (i.e., an estimator that includes a separate constant term for each governor to control for time-invariant confounders):

$$y_{i+t,j} = \gamma_1 Charisma_Speech_{i,j} + FE_j + \boldsymbol{Controls}_{i+t,j} + \varepsilon_{i+t,j} \quad (2)$$

FE_j designates the governor-specific constant term (fixed effect) in Table 1a, columns 2 and 7. The vector $\boldsymbol{Controls}_{i+t,j}$ includes governor characteristics (e.g., gender, age, education, party and tenure), state characteristics (e.g., GDP, population number and demographics, density and poverty). Because some variables, at least in part, are endogenously determined, we include controls only in some models to avoid a "bad control" problem (cf. Table 1a, columns 4, 5, 9 and 10). Hence, we can assess whether the estimated coefficients substantially change with respect to the models without these controls (Table 1a, columns 3 and 8, respectively).

Thereafter we replace FE_j with a vector of cluster means, $\boldsymbol{a}$, of all level 1 variables (i.e., that vary within cluster), add a cluster-specific error term, u_j, and use the random-effects GLS estimator (see Antonakis, Bastardoz, & Rönkkö 2021; McNeish & Kelley, 2018) in Table 1a columns 3, 4, 5, 8, 9, and 10):

$$y_{i+t,j} = \gamma_1 Charisma_Speech_{i,j} + \boldsymbol{a}_j + \boldsymbol{Controls}_{i+t,j} + \varepsilon_{i+t,j} + u_j \quad (3)$$

To give a basic sense of the governor-level correlation of the charisma signaling score with stay-at home (+1 day) see Appendix H; we include too a second correlation, where we have partialed out the control variables both from the charisma score and stay-at-home before hand (for the regressors in the full model). These correlations though ignore the within-level variability of governor charisma signaling, which is the core specification we are concerned about. Thus, we draw conclusions only from the full panel regression specification.

To examine our research question, we first report Table 1a, columns 5 and 10 models as baseline models in Table 1b. On this basis, we introduce the variable "conservative," (Table 1b, columns 1 and 5). As mentioned previously, it is critical that we satisfy the assumptions of using a difference score (see Edwards & Parry, 1993). Briefly, a difference score can only be used if the coefficients of the two variables constituting the difference score are equal and opposite in sign—of course, within sampling error, which requires a statistical test. Given that we estimated the variance using a cluster-robust estimator we used Wald tests, which are valid for such estimators (Wooldridge, 2002, pp. 290–291). We first compared the estimators for Models 1 and 5 in Table 1b (without the interactions), where we include the difference score or its two components. We then perform two tests (a) whether the coefficients are not significantly different when an equality constraint is imposed on them (making them equal and opposite in sign), and (b) whether the rest of the coefficients differ when including the difference score; this latter Wald test, referred to as a Chow (1960) test is a generalized Hausman-type test to ensure that including the difference score does not engender omitted variable bias (by changing the rest of the coefficients). This test goes one step beyond what Edwards and Parry (1993) suggest, ensuring the robustness of our results. In terms of test for (a) the difference in the coefficients for constituent terms of the difference score for both the 1-day and 7-day measures were not significant when we imposed the constraint: $\chi^2(1) = 1.95$, $p = .16$ and $\chi^2(1) = 1.61$, $p = .20$ respectively; for (b) the Chow tests showed no difference in the rest of the coefficients when including the variable "conservative" for both the 1-day and the 7-day measure: $\chi^2(31) = 20.91$, $p = .91$ and $\chi^2(31) = 13.33$, $p = .99$. Appendix I reports these results in full along with our procedure for implementing both tests.

Having established that a difference score is appropriate, next, we interact the difference-score variable with the charisma score (Table 1b columns 2 and 6). Note, we include too the interaction with the charisma score cluster mean (i.e., average charisma score by governor), as well as the three-way interaction as a precautionary means against the possibility that the unobserved fixed effect interacts with our level 1 or 2 regressors (see Bai, 2009). As a further precaution, we introduce the quadratic effects of the variables constituting the interaction in the event that they partly or wholly drive the observed

Table 1a
Predicting stay-at-home behavior from charisma of governor speeches.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Stay-at-home: 1 day after governor speech					Stay-at-home: 7-day average after governor speech				
Charisma	.11***	.05***	.05***	.04***	.06*	.11***	.06***	.06***	.05***	.07**
	(6.81)	(3.83)	(3.83)	(3.57)	(2.39)	(6.64)	(4.35)	(4.35)	(3.92)	(2.96)
Avg. charisma			.06**	.05**	.14***			.06**	.06**	.16***
			(2.82)	(2.73)	(3.74)			(2.80)	(2.70)	(3.84)
Days since 2/28/2020	.50***	.50***	.51***	.53***		.48***	.48***	.52***		
	(18.02)	(17.97)	(15.35)	(15.50)		(16.62)	(16.56)	(13.75)		(13.95)
No. cases			.02*	.02†				.02†		.01
			(2.26)	(1.86)				(1.73)		(1.25)
No. casualties			−.22*	−.17				−.17		−.11
			(2.05)	(1.60)				(1.46)		(.94)
No. days shelter-in-place order in place			−.13*	−.20**				−.22**		−.30***
			(2.18)	(3.43)				(3.23)		(4.39)
No. of sentences				−.01						−.02
				(.79)						(1.19)
Governor FE	No	Control cluster means for *ij* variables				No	Control cluster means for *ij* variables			
		Yes					Yes			
Month fixed effects	No	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Additional controls	No	No	No	No	Yes	No	No	No	No	Yes
R-squared	.17	.85	.85	.86	.86	.19	.80	.80	.82	.82

Note: The observation is a given speech *i* of a given governor *j*. The sample covers for all 50 US governors 7 speeches on COVID-19; $n = 350$. Significance level depicted by $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, $^{\dagger}p < 0.1$. Standard errors clustered at the governor level. *t*-statistics in parentheses. R-squares for panel estimators are for the within level. The variable Charisma is centered for models 3, 4, 5, 8, 9, and 10. Additional controls include governor characteristics: gender, age, education, party and tenure, and state characteristics: GDP, population number, population demographics (%black, %Hispanic, %obese, %over 65 years of age, % with at least bachelor's degree), density, poverty, and number of prior disasters.

effects (see Table 1b columns 2 and 6; Cortina, 1993). Finally, we include an interaction term with the political party of the governor, and estimate the full model (Table 1b, columns 4 and 8).

Results

Fig. 1 shows the distribution of the charisma scores for our 350 observations—seven speeches for each of the 50 governors (mean = 45.8; SD = 31.1). We depict both the raw score and the predicted score conditioned on a series of control variables including, number of sentences and words used (which, as mentioned, mechanistically determine the upper and lower boundaries of the charisma score), political party affiliation (Republican vs. Democrat), sex (male vs. female), inauguration date, education level (4 levels), attendance of Ivy League school, parent politician (to capture family legacy or learning effects), history of tragedy in the family (in case a personal tragedy affects governor discourse), and governor age. Alone, the number of sentences were strongly predictive of scores, $F(1, 48) = 711.18$, $p < .0001$, *r*-square = .94; the rest of the controls also added to the variance explained, $F(10, 38) = 1.94$, $p = .07$, *r*-square change = .02.

Interestingly, both distributions are significantly, positively skewed ($p = .004$ and .002, respectively), with lots of variation. These results suggest that charisma signaling is not that strongly displayed by many governors; perhaps they are unaware of its utility or aware of it but cannot easily display it or might not care to display it. Or they may be unaware of the specific communication tactics that help produce the signal but still know that charisma per se is important. Whatever the case, there is large variability in our measure of charisma signaling, which is ideal for estimation. Thus, we can assume that there is potentially little endogeneity in the charisma score by way of selection (given that many governors have low scores, and the distribution is not truncated or negatively skewed), increasing the confidence we can ascribe to our interpretation of the effect. We also note that the governor charisma score shows reasonably strong stability (Interclass correlation, *ICC1* = .51, 95% *CI* = .38–.63) and is measured very reliably (Reliability of mean, *ICC2* = .88), suggesting strong between-governor differences (though still with much within variation). Some governors "have it," others do not.

Table 1a shows the main results on how the charisma score is associated with stay-at-home behavior in the days following each speech. Column 1 depicts the raw relationship: A change in 10 units of governor charismatic signaling content in a particular speech is associated with an increase in stay-at-home behavior of 1.10 percentage points.

The result continues to hold once we filter out all time-invariant governor and US state characteristics (including a governor's average charisma score) with governor-specific constant terms (fixed effects) and add time fixed effects and the number of days lapsed since the first speech (February 28th, 2020) in column 2. The estimated coefficient is of a similar order of magnitude and remains statistically significant. Hence, the identification strategy in this column relies on exploiting deviations from usual charisma score levels. If, for example, a given governor usually comes across as rather dull but on a given occasion stages a grand charismatic performance, this charisma score is exploited for identifying how charisma signaling affects stay-at-home behavior of state residents on the following day.

Column 3 presents a similar specification, but instead of all time-invariant governor and state characteristics it controls for average charisma score of a given governor (see Mundlak, 1978); given the cluster mean centering of charisma scores means that the average charisma score captures the "between" effect (Antonakis et al., 2021). This method has the advantage of accounting for the fixed-effect of charisma signaling (the average charisma score) and not only charisma shocks (deviations from the mean charisma score). It also allows for the inclusion of the mean cluster values of key health-related controls (e.g., the number of infections) in columns 4 and 5, in addition to augmenting the specification with a further set of con-

Table 1b
Predicting stay-at-home behavior from charisma of governor speeches and political ideology.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Stay-at-home: 1 day after governor speech				Stay-at-home: 7-day average after governor speech			
Charisma (Ch)	.06*	.08†	.08†	.08†	.07**	.11**	.13**	.13**
	(2.37)	(1.80)	(1.75)	(1.95)	(2.94)	(2.84)	(3.33)	(3.41)
Average charisma (AvCh)	.10**	.10**	.18***	.12*	.11**	.11**	.19***	.13*
	(3.07)	(3.03)	(3.52)	(2.34)	(3.15)	(3.11)	(3.54)	(2.58)
Conservative (Con)	.06	.09†	.12*	.15*	.06	.09	.11†	.12
	(1.63)	(1.66)	(2.32)	(2.02)	(1.46)	(1.64)	(1.90)	(1.50)
Republican vote 2020	−.18***	−.18***	−.14**	−.16***	−.20***	−.20***	−.16**	−.18***
	(4.51)	(4.59)	(3.09)	(3.73)	(4.52)	(4.77)	(2.90)	(3.86)
Republican party (R)	−.61	−.63†	−.68†	−3.61*	−.64	−.66	−.83†	−4.38**
	(1.58)	(1.67)	(1.70)	(2.46)	(1.48)	(1.56)	(1.96)	(2.85)
Days since 2/28/2020	.53***	.54***	.53***	.54***	.55***	.56***	.54***	.55***
	(15.43)	(16.08)	(15.87)	(15.54)	(13.87)	(14.27)	(13.95)	(13.58)
Ch*Con		−.00	−.00	−.00		−.00**	−.00**	−.00
		(1.26)	(1.25)	(.22)		(2.78)	(2.86)	(1.01)
Ch*AvCh		.00	.00	.00		.00	−.00	−.00
		(.41)	(.37)	(.28)		(.09)	(.23)	(.38)
Con*AvCh		−.00	−.00	−.00†		−.00	−.00	−.00
		(.81)	(1.38)	(1.82)		(.95)	(1.52)	(1.38)
Ch*Con*AvCh		−.00	−.00	−.00		.00	.00	−.00
		(.72)	(.73)	(.88)		(.16)	(.10)	(.29)
R*Ch				−.29*				−.22*
				(2.23)				(2.06)
R*Con				.04				.09
				(.56)				(1.15)
R*Ch*Con				.01				.01
				(1.56)				(1.36)
R*AvCh				.07*				.09*
				(1.97)				(2.42)
R*Ch*AvCh				.01*				.01*
				(2.28)				(2.05)
R*Con*AvCh				−.00				−.00
				(.62)				(1.36)
R*Ch*Con*AvCh				−.00†				−.00
				(1.84)				(1.59)
Con*Con			−.00	−.00*			−.00	−.00
			(1.30)	(1.97)			(.86)	(1.33)
Ch*Ch			−.00	−.00			−.00**	−.00**
			(.16)	(.19)			(2.80)	(2.83)
AvCh*AvCh			−.00†	−.00			−.00	−.00
			(1.71)	(.61)			(1.40)	(.45)
No. cases	.02†	.01†	.01†	.01	.01	.01	.01	.01
	(1.87)	(1.68)	(1.68)	(1.61)	(1.26)	(.97)	(.98)	(.93)
No. casualties	−.17	−.14	−.14	−.14	−.11	−.08	−.08	−.08
	(1.60)	(1.56)	(1.57)	(1.47)	(.94)	(.77)	(.79)	(.74)
No. days SIP in place	−.20**	−.20**	−.20**	−.20**	−.30***	−.31***	−.29***	−.29***
	(3.41)	(3.44)	(3.45)	(3.38)	(4.36)	(4.38)	(4.31)	(4.16)
No. of sentences	−.01	−.01	−.01	−.01	−.02	−.01	−.01	−.02
	(.79)	(.80)	(.80)	(.86)	(1.18)	(1.20)	(1.14)	(1.20)
Governor FE	Control cluster means for *ij* variables				Control cluster means for *ij* variables			
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	.86	.87	.87	.87	.82	.84	.84	.84

Note: The observation is a given speech *i* of a given governor *j*. The sample covers for all 50 US governors 7 speeches on COVID-19; $n = 350$. Significance level depicted by $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, $^{\dagger}p < 0.1$. Standard errors clustered at the governor level. *t*-statistics in parentheses. R-squares for panel estimators are for the within level. The variable Charisma is centered for all models.

trols to ensure the estimator is consistent. Note, the addition of the cluster means ensures we have results emulating the fixed-effects estimator (see also McNeish & Kelley, 2018). Finally, columns 6-10 replicate the first five columns, but focus on the effect on the week average —that is, days 1 through 7—after the given speech. Overall, the estimated coefficient of the impact of charisma signaling in a given speech is in all columns statistically significant at the 1 percent level with a coefficient that is relatively stable and practically important. The above results largely support Hypothesis 1.

In Table 1b we investigate the core research question of how the charisma score interacts with political ideology. For this purpose, we study how the effect of the charisma score of a given speech, as well as average charisma score, matter differentially for conservative or Republican voters. Looking at average marginal effects (i.e., $\partial y/\partial x$), it is interesting to observe that Republican vote share is negatively related to stay-at-home behavior across all specifications. Although those states where most voted for Republicans are less likely to distance it is encouraging to note that the within and between average marginal effects of charisma signaling are always significantly related to the outcome. To examine potential moderating effects, the highest level of interaction that was significant for both the 1-day and 7-day stay-at-home behavior measure is that of the three-way interaction: Republican*Charisma*Average Charisma. What this interaction means is that a particular speech that has a high score by a Republican politician has a stronger impact when the average charisma score of this politician is larger (i.e., and hence maybe the current speech receives

Table 2
Coded charismatic leadership tactics in treatment vignette.

		Charismatic Leadership Tactic (CLT)								
No.	*Sentence*	1	2	3	4	5	6	7	8	9
1	This is not a sprint, my friends, this is a marathon.	1			1					
2	You have to gauge yourself.									
3	And even though it's so disruptive, and so abrupt, and so shocking, it's also long-term.					1				
4	Stay six feet away from people.									
5	Don't be reactive, be proactive.				1					
6	There's economic anxiety, people are out of work.									
7	What does this mean?		1			1[a]				
8	Unemployment insurance?		1							
9	Will it cover the bills?		1							
10	This fear of the unknown, this misinformation.									
11	You put it all together, it is very disorienting, to say the least.							1		
12	If you are feeling disoriented, it's not you, it's everyone, and it's everywhere, and it's with good cause.				1	1		1		
13	Also, we have to plan forward on testing.									
14	We've mobilized, we've scrambled, but this is still not where it needs to be.									
15	We need many more tests.									
16	The social distancing is important.									
17	You don't win on defense, you win on offense.	1			1					
18	Don't get complacent.									
	Total	2	3	0	4	3	0	2	0	0

Notes. Table based on Antonakis 2017, p. 74. CLT1 = metaphor, CLT2 = rhetorical question, CLT3 = story, CLT4 = contrast, CLT5 = list, CLT6 = moral conviction, CLT7 = sentiment of the collective, CLT8 = ambitious goal, and CLT9 = goal can be achieved. [a] List begins and runs over into sentences 8 and 9. The correlation between human and computer coding at the total (individual) CLT level is $r = .83$ (.73). The concordance correlation coefficient, which accounts for mean difference too at the total (individual) is rho = .82 (.68). Regressing the human coding on the computer coding gave a coefficient of *beta* = 1.06, $SE = .06$, $t = 17.49$, $p < .001$; this estimate demonstrates an almost one-to-one concordance of the computer coding with respect to the human coding. The estimate did not change significantly when controlling for CLT type.

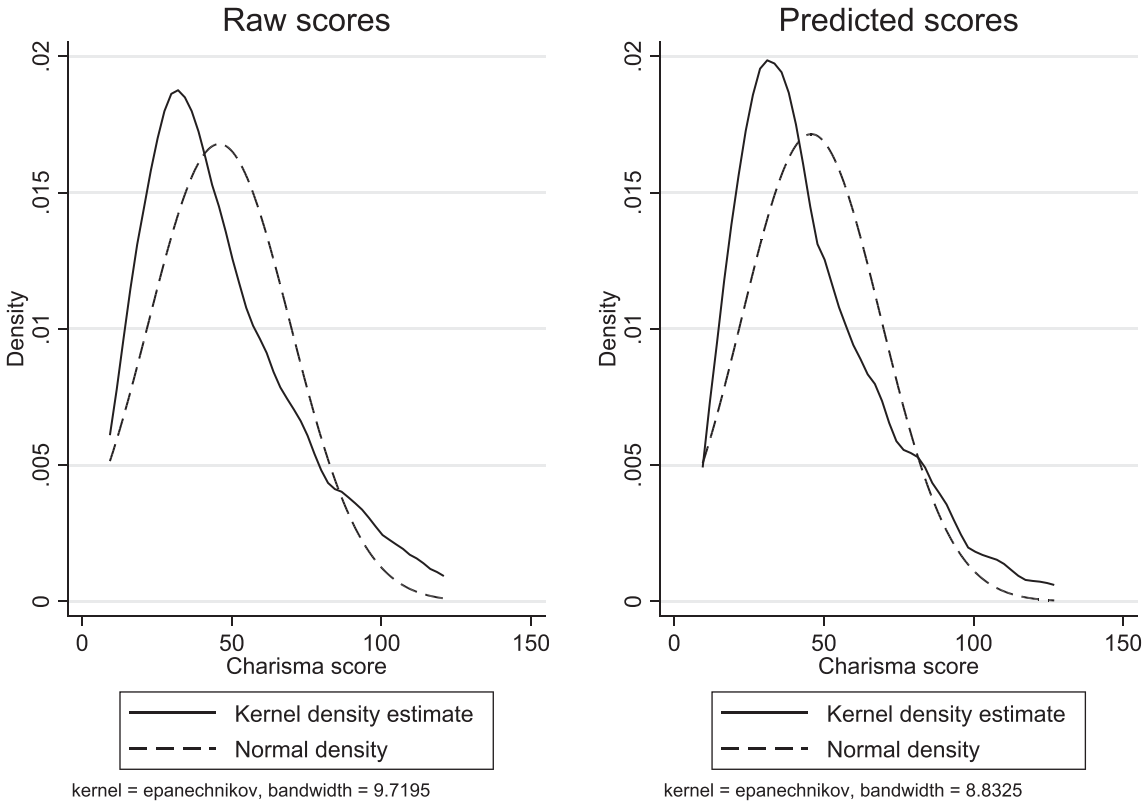


Fig. 1. Distribution of governor average charisma scores.

more attention). We plot this result for the 7-day average outcome as a function of governor party affiliation at +1 and −1 *SD* from the mean for the other two covariates. Given the significance of the interaction, the four slopes are different from each other.

As Fig. 2 shows, the overall effect for charisma signaling and the average charisma score is positive. The steepest slope is that of Republican governors at a high level of charisma scores ($\beta = 11.79$, $SE = 2.37$, $z = 4.98$, $p < .0001$). The rest of the slopes are all signif-

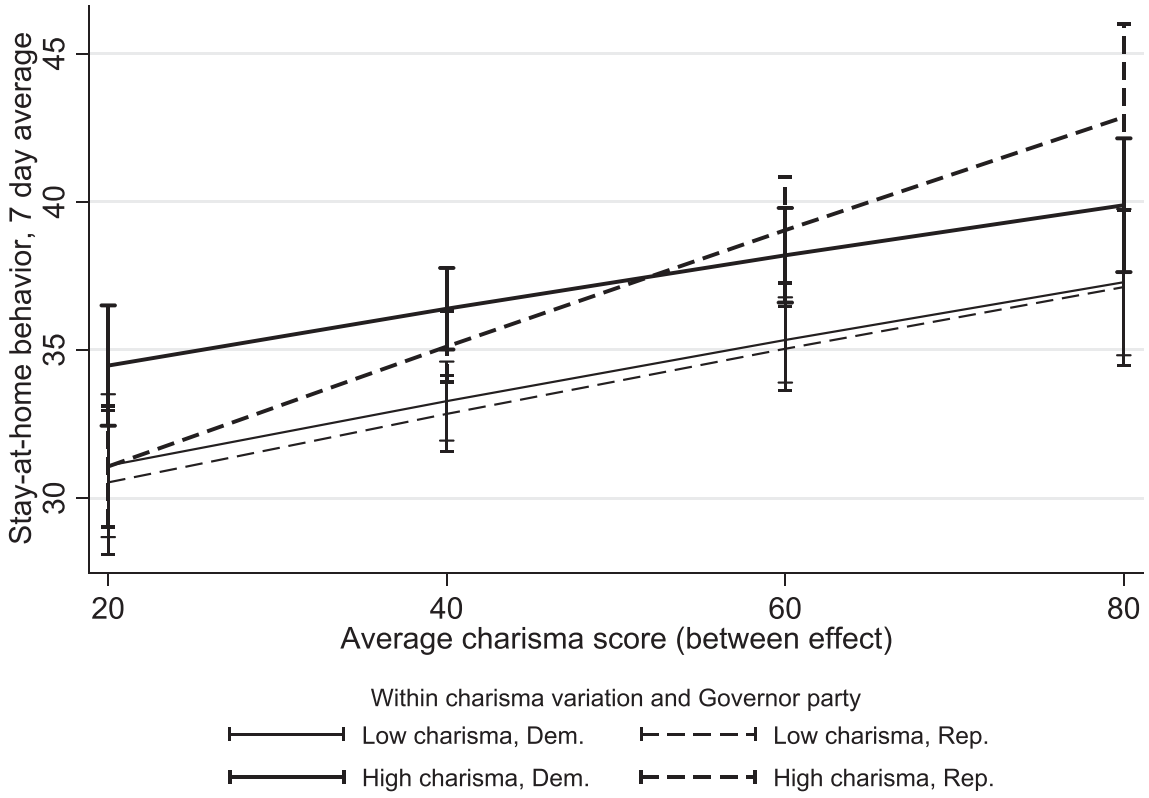


Fig. 2. Moderation of governor party on charisma's effect on percentage of people staying at home (7-day average).

icant at $p < .01$ with coefficients of 6.19 (Democrat, low charisma score), 6.59 (Republican, low charisma score), and 5.41 (Democrat, high charisma score). Interestingly, the slope for Republican governors with a high charisma score is significantly higher than that of Democrat governors with a high charisma score, $\chi^2(1) = 9.94$, $p = .002$ (we got a similar result for the 1-day average outcome, $\chi^2(1) = 6.63$, $p = .01$). Thus, there appears to be a moderating effect for governor partisanship.

However, the variable conservative did not play a consistent and significant moderating role in the two specifications. This result is probably not attributable to multicollinearity between political ideology (conservatism) and governor partisanship, given that these variables are not that highly correlated at the state level, $r(50) = .45$, $p = .001$ (suggesting about 20% shared variance).

Brief discussion

Given the limitations (noted below) and predicated on replication with a stronger causal design, the above results can have important practical implications. To determine the importance of charisma signaling per se, we refer the reader back to the first coefficient reported in Table 1a; .11. This result implies that a one-point higher charisma score in the speech reduces mobility by .11 percentage points. The charisma measure features a mean of 45.8 and standard deviation of 31.1. Hence, a one standard deviation higher charisma score (31 points) leads to roughly 3.3 percentage points higher stay-at-home behavior on the next day (which corresponds to 10% of the baseline level of stay-at-home behavior —this variable has a mean of 33.7 and a standard deviation of 8.3). From Wilson's (2020) coefficients of Table N1, one can compute the change in COVID-19 fatalities with respect to mobility, which is around .61, cumulatively over a 10-week period. Thus, a 10% increase in stay-at-home behavior would result in a 6.1% decrease in COVID-19 fatalities. This estimate is epidemiologically important as illustrated by the order of magnitude of total fatalities from COVID-19.

Focusing on the time frame covered in our study (02/28/2020–04/23/2020), and adding three weeks from the day of the last observed speech as this represents the typical, average reported length from manifestation of symptoms to death (i.e., 05/14/2020), according to our estimates a one standard deviation higher charisma score in governors could potentially have saved 5,350 lives in the United States (6% of 89,179; Dong et al., 2020) of that period. Even if the effect of charisma signaling was halved it could still reduce deaths by an important amount (i.e., even for the coefficient of column 5 with the full array of controls, the numbers of lives saved would be more than 2, 675 during the time frame of the study, and over 29,840 over the course of the pandemic).

Of note is that the above rather approximate calculation only looks at the within effect; considering the between-governor differences shows that the total effect (i.e., the sum of the within and contextual effect, see McNeish & Kelley, 2018; Antonakis et al., 2021) is very large. Using the estimate of the between effect (mean charisma score) from column 5 in Table 1a suggests a governor having an average charisma score that is one-point higher reduces mobility by .19 percentage points, which is about twice as much as the within effect.

It is of course important to not overinterpret this rough computation, because there is considerable heterogeneity in how the virus unfolds in different regions and with different populations; the effect of charisma signaling may well be zero in various contexts and for various groups. Interestingly, we found that the effect of charisma signaling (i.e., the average charisma score of the governor and the charisma score of a particular speech) is moderated by governor partisanship. However, we should recognize that this result perhaps hides some intricacies that we cannot properly evaluate. Perhaps the persona of the governor plays an important role here, independent of party affiliation and whether the state is predominantly politically conservative or voted Republican in the last presidential election. Indeed, the corre-

lations between being politically conservative or politically liberal and whether the governor is Republican are $r = .45$ (i.e., about 20% shared variance) and $r = -.43$ (i.e., about 18% shared variance) respectively, suggesting that there is variation in the degree of conservatism in states and that, regardless of the party affiliation of the governor, individuals care more about the specific ideology of governors (Lelkes, 2021). For instance, Arnold Schwarzenegger had quite broad-based support when he was governor of California.

Yet, perhaps there is still a confounding factor at play because some citizens may not cooperate with a governor's request given that citizens are not aligned with the political party the governor represents, or perhaps citizens may not appreciate the governor for other reasons. Thus, it is not possible for us to hold constant factors that may matter in citizens' decision to maintain distance and stay at home.

To better understand the dynamics political ideology (conservatism) may play for willingness to stay at home, it is crucial to expose individuals to speeches wherein the identity of the governor is unknown. This design is needed so that we can better understand the causal effect of charisma signaling, independent of governor characteristics.

Study 2

Method

Procedure

Study 2 tests the causal impact of anonymous governors' communication with higher or lower charisma signaling on incentivized choices related to beliefs about distancing and willingness to stay at home using a three-group between-subjects randomized vignette experiment. A priori power calculations for a one-way ANOVA F-test with three groups, an expected effect size of about .23 (R^2-change = .05), error rate of .05, and power of .80 yields a minimum sample size of 190. We decided to triple this number to err on the side of caution and to ensure we would have sufficient power for heterogenous analyses along participants' political ideology. To recruit participants, we used Amazon's Mechanical Turk (Mturk); an online labor market platform commonly used by social science researchers for survey and experimental research. We compensated participants a flat fee of $.75, which was calculated based on an anticipated average completion time of 6 minutes and the current federal minimum wage of $7.25. On December 12th, 2020, 740 adults residing in the United States were recruited. We had 685 participants with 661 individuals providing complete responses on the survey, making up our final sample. Appendix J details the flow of the survey as well as attrition numbers.

Manipulation

Speech vignettes

We manipulate governor charisma signaling by strategically presenting excerpts from real—but anonymized—COVID-19 press briefing speeches that use many or few charismatic leadership tactics. Using information from actual leadership communications ensures high contextual realism of the communication scenarios and aligns with recommendation to make vignettes as realistic as possible (Lonati et al., 2018). Using the data from Study 1, we identified prototypical examples of speeches with high and low charisma scores, and pulled excerpts from Andrew Cuomo's March 28, 2020 COVID-19 press briefing and Kim Reynolds' March 20, 2020 COVID-19 press briefing. Note the mean charisma score for a governor's speech was 45.80. Cuomo, with a mean of 106.35 across his speeches, had the second highest mean charisma score among the governors; Reynolds had one of the lowest scores (i.e., 23.06). This comparison is fair and does not engender diametrically different treatments that would confound effects. We are comparing a speech with a low charisma score (i.e., baseline/control) to a speech with a high charisma score. Having different "dosages" of charisma thus ensures that we do not fall in the trap of having a "poison vs. medicine" problem (Lonati et al. 2018).

We culled excerpts of a speech (a) of Andrew Cuomo, which had a high charisma score, (b) originating from less charismatic parts of the same Cuomo speech, and (c) emanating from a very low scoring speech by Governor Kim Reynolds. Having a Cuomo control speech, coupled with another control speech thus ensures there is no "Cuomo effect"[2] present beyond the use of charisma. On this basis, we compiled excerpts to create three equal-length speeches. The speeches are depicted in Appendix K. Importantly, the speeches are of similar length: 143, 142, and 143 words, respectively and, as readers can verify, the speeches make similar informational demand effects on the participants. Moreover, we ensure that no identifying information would be in the speeches so that they remained anonymous to the participants.

Manipulation checks

To ensure that our three vignettes indeed manipulate charisma, we performed three a priori manipulation checks. First, we applied our computer algorithm to each of the speeches. Our treatment condition created from Cuomo's March 28 speech yields a total charisma score of 11.73, with the two control conditions yielding scores of 4.68 (based on Cuomo's March 28 speech) and 2.99 (based on Reynolds' March 15 speech).

Second, we used a very experienced and trained human coder, unaware of the purpose of the study and its conditions to code the speeches for the absolute presence of a charismatic leadership tactic. The coder gave the following initial respective total scores per speech: 15, 2, 0. The correlation at the speech level (9 tactics * 3 speeches) between the coder and the computer algorithm was $r(27) = .71$, $p < .0001$. The concordance correlation coefficient between coder and computer was .73, showing substantial agreement (Lin, 1989). The codings were also discussed with the last author of the study to verify their validity resulting in the following respective scores: 14, 2, and 0. Table 2 shows coding of the Cuomo charismatic condition to illustrate the communication tactics used to engender charisma and to showcase how we quantified leader charisma based on the press briefings. Beyond validating the manipulation, these results provide strong validity for the use of the algorithm used to rate the speeches in Study 1.

Third, we performed an external manipulation check on an independent sample of 223 participants recruited via MTurk on November 2nd, 2020. Individuals were randomized to one of our three vignettes. After reading the vignette, participants were asked to rate how (a) inspiring and (b) charismatic they believed other people would find the speech they just saw. Each rating was completed on a scale ranging from 1 ("very uninspiring"/"very non-charismatic") to 5 ("very inspiring"/"very charismatic"). Appendix L outlines details of the manipulation check, including data collection and measurement. In line with our expectation, participants exposed to the high-charisma Cuomo condition found the speech more inspiring and charismatic compared to participants exposed to either of the low charisma conditions. This result is important because it shows that the objective difference in charisma signaling was conveyed in the minds of human subjects, resulting in the charismatic effect (thus we do refer to "charisma" per se when relevant). Finally, masking the sender of the speeches makes the conditions subtler and removes any person-specific effects of the messenger.

[2] Speech selection and the experiment occurred prior to sexual harassment allegations being made against former New York Governor, Andrew Cuomo. Criminal charges, however, have since been dismissed. Still, we would have preferred to have selected another case had we know of the events that transpired.

Measures

Stay-at-home beliefs and willingness

Following exposure to one of the conditions, we elicited two measures on the belief about and willingness to engage in stay-at-home behaviors. It is important to note that neither of these measures, unlike that employed in Study 1, capture observed, real-world behavior. First, we asked participants about their preference for two mitigation behaviors that closely resemble stay-at-home behavior, namely how *willing* they would be to (a) avoid crowded places like restaurants, bars, or sporting events, and (b) not attend family events. Responses were provided on a scale from 0 (not at all willing) to 10 (completely willing). Second, we used a "coordination game" approach to elicit participants' belief about how likely other people on average would be to conduct the same two mitigation behaviors using the same scales. For each mitigation behavior, we told participants they could earn up to $.25 based on the accuracy of their estimate for a total bonus incentive of $.50. This incentive is 66.67% of the flat fee for completing the survey and represents a high-powered incentive. Using this second, "incentivized" measure serves the purpose of addressing reporting bias. If there are no incentives involved for answering survey questions, respondents may give socially desirable responses or simply not be motivated to give a thoughtful response. In contrast, when facing incentives to estimate overall expected behavior, participant replies should feature more accurate estimates and lower reporting bias. Like other social behaviors, distancing has a strong element of coordination and conditional cooperation (Frey and Meier, 2004). It is thus reasonable to expect a positive correlation between the beliefs about others' willingness to engage in stay-at-home behaviors and participants' own actual stay-at-home behavior. Given that both measures are endogenous, and because we have a manipulated instrumental variable, we estimate how these variables relate via instrumental-variable estimation (Sajons, 2020).

For our main analyses, we aggregate participants' two estimates of mitigation behaviors to a general mean estimate of participants' willingness to stay at home. A similar pattern of results emerges when the two measures are analyzed separately, but we report the index because it is broader and more reliable.

Political ideology

As part of our survey, participants were asked to describe the extent to which they identify as "extremely liberal," "liberal," "slightly liberal," "moderate; middle of the road," "slightly conservative," "conservative," or "extremely conservative." Participants were also provided the option "I haven't thought much about this," which we code as missing for the purpose of our analyses. We use this scale in its original form to denote the degree of "political conservatism" on a 7-point scale, and in a condensed form where we collapse the "liberal" and "conservative" response options to create 3 groups (i.e., liberal, moderate or conservative).

Covariates

Asking participants to estimate the mean response of a group is a cognitively demanding task; thus to obtain a measure of their cognitive ability we therefore asked them to complete a modified version (Jordan et al., 2021) of the Cognitive Reflection Test, CRT (Frederick, 2005). The CRT is "a simple measure of one type of cognitive ability" (Frederick 2005, p. 26), and has been shown to predict heuristic decision making, and correlate with working memory capacity and established measures of general cognitive ability (Frederick 2005; Toplak et al., 2011). Frederick (2005), for instance, reports a correlation of $r = .43$ between CRT scores and the Wonderlic IQ test, suggesting a moderate overlap with cognitive ability. For sake of brevity, we label this variable "CRT score," recognizing that a comprehensive measure of individual intelligence would encompass facets not captured here. Specifically, the "CRT score" denotes the number of correct answers to each of the following questions: (a) *The ages of Mark and Adam add up to 28 years total. Mark is 20 years older than Adam. How old is Adam?* (b) *If it takes 10 seconds for 10 printers to print out 10 pages of paper, how many seconds will it take 50 printers to print out 50 pages of paper?* and (c) *On a loaf of bread, there is a patch of mold. Every day, the patch doubles in size. If it takes 40 days for the patch to cover the entire loaf of bread, how many days would it take for the patch to cover half of the loaf of bread?*

Because a participant's estimate of the mean of the group should, theoretically, be predicted by this CRT score, we create a variable called "Over rater" (=1 if the participant's score on the incentivized belief measure was above the mean of the group; else = 0), which we interact with the CRT score (Edwards, 1995). Thus, the smarter the individual is, the closer to the mean the individual will be as a function of the CRT score regardless of whether they are an over or under rater. Results (Table 3b) show the expected pattern (i.e., a significant and negative interaction effect). Results for the key variables were very stable, though less significant when not modeling the congruence measure (from CRT scores and over-under rater) given that estimator has more error—that is, is less efficient and hence increases the standard errors of estimation.

We incorporated several other covariates, including participants' gender (1=man, else =0), their age as a continuous variable, and their state-level geolocation. Chi-square tests indicate that assignment to experimental group (whether we use the original three treatment group assignment or collapse the control conditions), does not predict these covariates (lowest p value = .44). These results bolster our confidence that the randomization was successful in creating ex ante identical groups.

Attention checks

We also included three attention checks at the end of the survey. One question asked participants to recall whether the governor asked residents to physically distance whereas the other two were factual recalls of minimum time for hand washing and minimum distance to others as prescribed by a CDC video. Conditioning the sample on correct recall of all three checks generally reproduces the main pattern of results, with one exception (the coefficient for the interaction term between the treatment dummy and conservative orientation in model 6, Table 3a is insignificant, but the estimate is not significantly different from that of the full sample size, $\chi^2(1) = .03$, $p = .86$. However, given the reduced sample size, and the fact we do not include over-under rate and CRT score suggests this estimator is underpowered). We did include Model 6 in Table 3a to show how estimates differ when excluding these two controls and the key parameters did not change from that of Model 5. Table M1 in Appendix M provides an overview of all study variables for the lab experiment. Table N1 in Appendix N reports the correlation matrix.

Estimation

To examine the effect of beliefs on distancing and willingness to stay at home, we used an instrumental-variable procedure. This procedure aims at purging possible endogeneity bias by harnessing the exogeneity of the manipulated variables, which—to the extent that the model does not violate the assumptions of the estimator—make for suitable instruments (Sajons, 2020). Specifically, we estimated, for participant *i*:

$$y_1 = \beta_0 + \beta_1 NoCharisma + \beta_2 Conservative + \beta_3 NoCharisma * Conservative + \boldsymbol{Controls} + e \quad (4)$$

$$y_2 = \delta_0 + \delta_1 y_1 + \delta_2 NoCharisma + \delta_3 Conservative + \boldsymbol{Controls} + v \quad (5)$$

where y_1 is participants' belief about what others would do and y_2 is willingness to stay at home. Given that we incentivized individuals to predict the average willingness of other participants to stay at home,

Table 3a
Experimental results: Effect on belief about others' willingness to stay at home.

	(1)	(2)	(3)	(4)	(5)	(6)
Male	.08	.07	.13	.08	.07	.14
	(.86)	(.77)	(.96)	(.83)	(.77)	(.99)
Age	.01†	.01†	.02***	.01†	.01†	.02***
	(1.93)	(1.76)	(3.85)	(1.89)	(1.90)	(3.99)
Cognitive Reflection Test (CRT)	.04	.04	−.16*	.02	.02	−.15†
	(.39)	(.41)	(2.00)	(.23)	(.23)	(1.96)
Over rater	2.83***	2.84***		2.83***	3.08***	
	(9.91)	(8.02)		(8.02)	(11.70)	
CRT*Over rater	−.20†	−.21†		−.19†	−.19†	
	(1.84)	(1.85)		(1.74)	(1.73)	
Ideologically conservative (IC)[a]	−.04	**.05**	**.08**	.05		
	(.80)	(.77)	(1.16)	(.83)		
IC*Over rater	.06	.06		.06		
	(1.07)	(1.20)		(1.12)		
Cuomo non-charismatic (CNC)		**.23**	**.24**			
		(.74)	(.63)			
Reynolds non-charismatic (RNC)		**.46**	**.79***			
		(1.48)	(2.12)			
CNC* IC		**−.12**†	**−.06**			
		(1.93)	(.59)			
RNC* IC		**−.13***	**−.22***			
		(2.31)	(2.46)			
CNC*Over rater		.04				
		(.18)				
RNC*Over rater		−.09				
		(.39)				
Non-charismatic[b]				.34	**−.15**	**−.04**
				(1.21)	(.83)	(.26)
Non-charismatic*IC				−.12*		
				(2.38)		
Non-charismatic*Over rater				−.02	−.03	
				(.11)	(.13)	
Conservative[c]					**.09**	**.14**
					(.74)	(1.02)
Non-charismatic*Conservative					**−.29****	**−.29**†
					(2.61)	(1.75)
Conservative*Over rater					.19†	
					(1.76)	
State fixed effects	Incl.	Incl.		Incl.	Incl.	
R-squared	.64	.65	.04	.65	.65	.03

Note: Dependent variable: Incentivized beliefs on others' willingness to stay at home. Heteroscedastic robust *t*-statistics in parentheses. $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, $^{\dagger}p < 0.1$; $n = 661$; baseline treatment condition for all models is Cuomo charismatic; [a, b, c] See notes in Table 3b. The Wald test for the joint difference of the bolded coefficients between Models 3 and 4: $\chi^2(3) = 6.59$, $p = .086$, and Models 5 and 6: $\chi^2(3) = .61$, $p = .90$.

and to reduce the error variance and hence improve the efficiency of the estimator, we controlled for their CRT score (smarter people should do better) and being an over or under rater (i.e., using a dummy variable to indicate if they were over, 1, or under, 0 rater). We interacted this dummy variable with the CRT score to capture the fact that as one is smarter, one gets closer to the mean (and thus correctly modeling discrepancy data of this structure, see Edwards, 1995), as well as with conservatism and the treatments. Finally, we controlled for geographic effect by including state-level dummy variables, as well as individual level controls like participant sex and age. ν and e are error terms.

Results

Table 3a indicates that being exposed to the charismatic treatment impacts beliefs about how likely others would be to distance and stay at home among individuals who identify as politically conservative. We present different estimators to show the robustness and stability of this result. We first include the demographic controls, being an over rater, the CRT and being ideologically conservative (Model 1); then we add the experimental manipulations and their interactions with being ideologically conservative (Model 2). In Model 3 we show that the pattern of results is stable when excluding the variable "over rater." In Model 4 we present results using a parsimonious model, where we collapse the two non-charismatic (conditions); this decision was guided by a Wald test performed on the specification of Model 2 where we tested this constraint: $F(3, 600) = .37$, $p = .77$. In Model 5, we present results of a more parsimonious model, where we pool together all liberals and we do likewise with all conservatives. Results remained very similar. Finally, in Model 6 we show the parsimonious results without the over rater and CRT variables; the interaction pattern remains unaffected, though this estimator is underpowered (given that we remove two important regressors that soaked up much of the unexplained variance).

We graph the marginal effects of charisma in Figs. 3 and 4. Compared to the most charismatic speech—the treatment of the two less charismatic speeches results in a significantly lower belief that others will distance and stay at home among participants who identify as politically conservative; however, there is no effect for participants identifying as politically liberal (and note that there would also be no overall effect of charisma when not moderated by ideology). Put differently, politically conservatives are more easily convinced that others will engage in mitigation behaviors after charismatic speeches. Individuals identifying as politically liberal are unaffected by charismatic communication. For example, results from model 5 show that there are significant differences between the treatments for individuals who identify as conservative ($p = .039$); however, not for individuals identifying as moderate ($p = .41$) or liberal ($p = .54$).

These results clearly highlight that charisma indeed fuels beliefs among politically conservatives—who are often COVID-19-sceptics—that others would engage in distancing and stay-at-home behaviors

Table 3b
Experimental results: Effect on own willingness to stay at home.

	(1)	(2)	(3)	(4)	(5)
Male	−.30†	−.29†	−.30†	−.31†	−.40*
	(1.85)	(1.77)	(1.85)	(1.92)	(2.53)
Age	.03***	.03***	.03***	.03***	.03***
	(4.40)	(4.42)	(4.38)	(4.09)	(4.08)
Cognitive Reflection Test (CRT)	−.03	−.05	−.03	−.03	−.00
	(.31)	(.45)	(.30)	(.29)	(.04)
Cuomo non-charismatic (CNC)		.04			
		(.09)			
Reynolds non-charismatic (RNC)		−.22			
		(.62)			
Ideologically conservative (IC)[a]	−.37***	−.39***	−.39***		
	(7.80)	(5.19)	(5.20)		
CNC*IC		.02			
		(.20)			
RNC*IC		.03			
		(.25)			
Non-charismatic[b]			−.09	**−.03**	**.03**
			(.28)	(.17)	(.17)
Non-charismatic*IC			.02		
			(.25)		
Conservative[c]				**−.67*****	**−.65*****
				(3.95)	(4.00)
Non-charismatic*Conservative				**−.13**	**−.13**
				(.62)	(.64)
State fixed effects		Incl.***	Incl.***	Incl.***	
R-squared	.18	.19	.18	.18	.12

Note: Dependent variable: Participants' own willingness to stay at home. Heteroscedastic robust *t*-statistics in parentheses $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, $^{\dagger}p < 0.1$; $n = 661$; baseline treatment condition for all models is Cuomo charismatic; [a]scores range from 1 (extremely liberal) through the midpoint 4 (moderate; middle of the road) to 7 (extremely conservative); [b]for parsimony, the two non-charismatic conditions pooled together; [c]for parsimony, scores of all liberals pooled together, as were those of all conservatives and thus they range from −1 (liberal) through the midpoint (0) to 1 (conservative). The significance levels for state fixed effects refer to the significance of the joint Wald test. The Wald test for the joint difference of the bolded coefficients between Models 4 and 5 is $\chi^2(3) = 1.39$, $p = .71$.

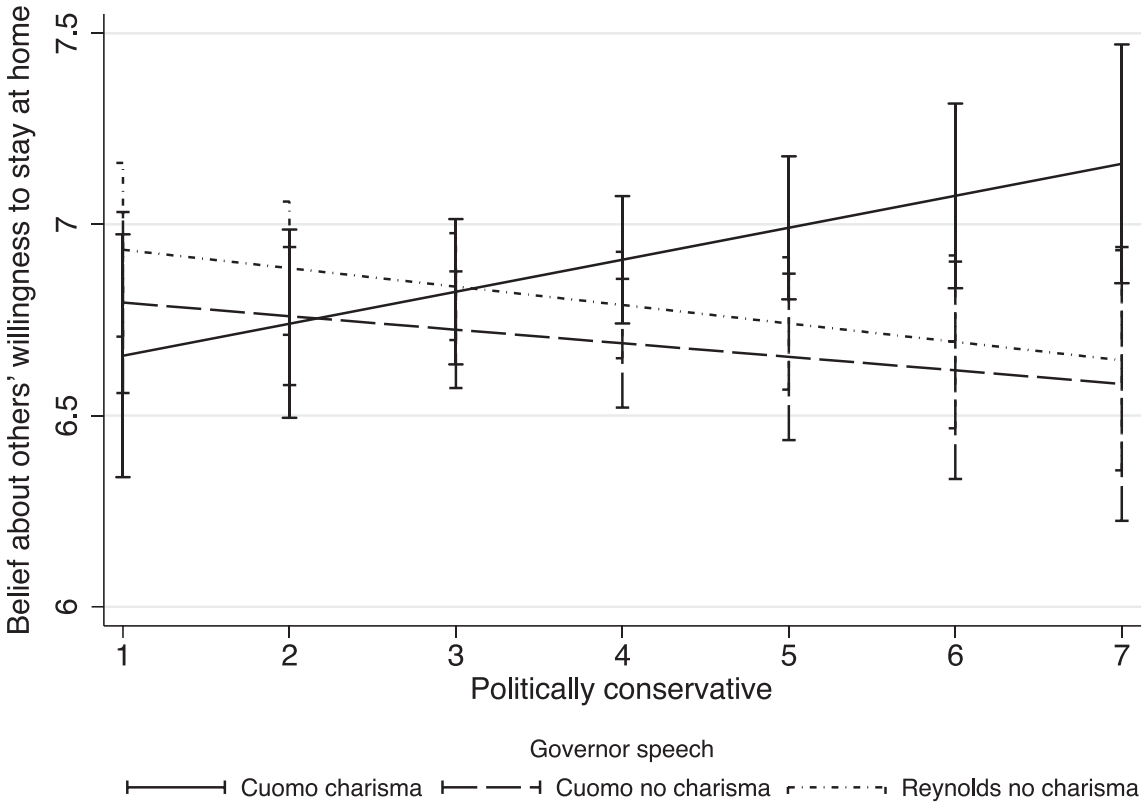


Fig. 3. Effect of charisma on beliefs.

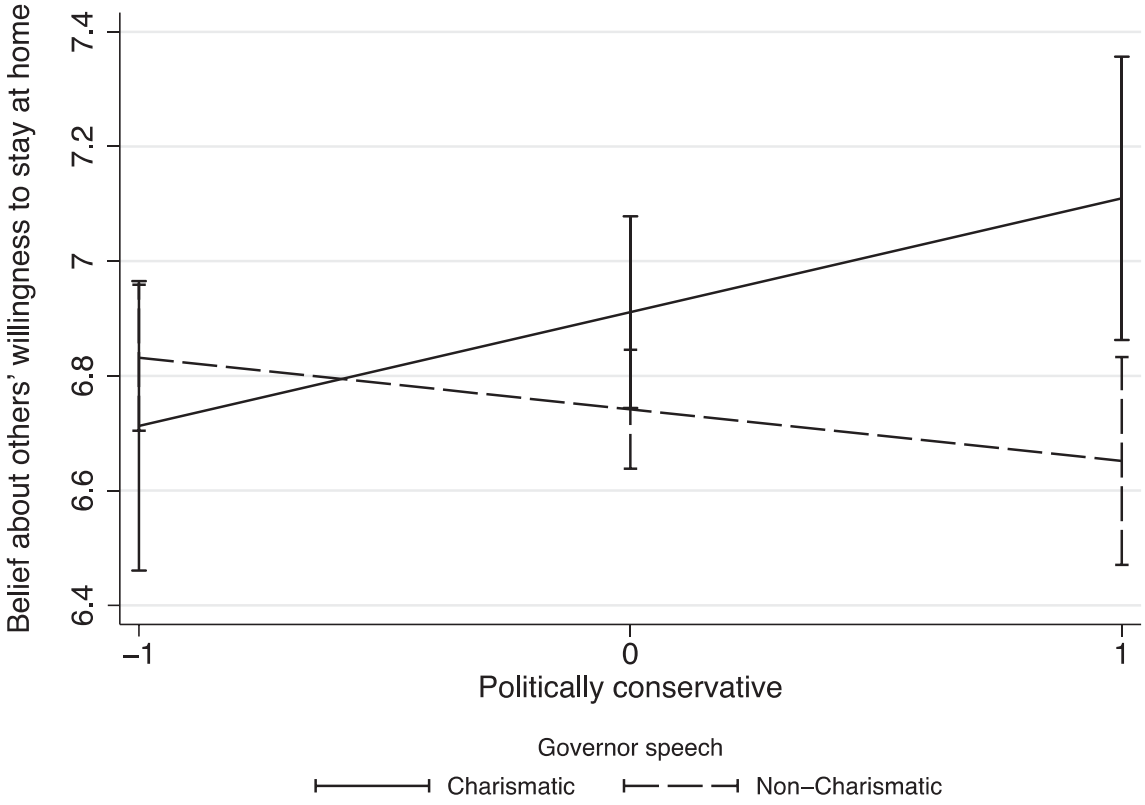


Fig. 4. Effect of charisma (uncharismatic treatments combined) on beliefs (ideology collapsed in three groups).

after hearing a charismatic speech; however, politically liberals who on average have a higher propensity to support anti-COVID-19 public health measures anyway (Gollwitzer et al., 2020) do not immediately react to charisma. Our results hold regardless of the specification used. Moreover, this result is not driven by differential in-group prediction; that is, when setting the reference mean to conservatives, moderates, or liberals, the main pattern of results holds. Thus, what expectations conservatives have are valid for all three groups.

Interesting too, controlling for gender, age, and fixed effects of education level (seven categories of educational attainment) of participants and state fixed effects, shows there is a negative relation (*coef.* = −.04, *SE* = .02, t = 2.18, p = .03) between participants' CRT score and their conservatism score. The standardized partial coefficient is statistically significant, but not very strong, −.09 (or −.12 disattenuated for unreliability, assuming .80 reliability in criterion and outcome), but present nonetheless (whether we used the long or short conservatism measure). These results follow previous large-scale studies using more complete measures of intelligence showing a negative correlation between intelligence and conservatism (Stankov, 2009).

Finally, Table 3b examines the impact of beliefs about how others will distance on participants' own willingness to stay at home. We see these results as more suggestive, because the behavior variable is not incentivized and may just reflect "cheap talk" (i.e., reporting bias may arise if participants tend to simply state what they believe others "want to hear"). Still the instrumental-variable procedure may remove this bias, though we can never be certain given that the estimator is only consistent if the instruments are the true ones for the endogenous regressor and the mechanism only works via the endogenous regressor.

First, participants interestingly reported themselves as more willing to engage in distancing and stay-at-home behaviors (mean response = 8.24, *SE* = .08) than they believed others would be (mean response = 6.79, *SE* = .07), t = 15.30, p < .0001. Second, we study whether—in a reduced form specification—the charismatic treatment directly predicts willingness to distance and stay at home (see Table 3b). The treatments had no effect; only conservatism (political ideology) did, whether it was coded on a 7-point (from 1 extremely liberal to 7 extremely conservative) or 3-point scale (from -1 liberal to 0 moderate, to 1 conservative). Individuals who reported identifying as more politically conservative indicated they would be less willing to engage in distancing and stay-at-home behaviors.

However, third and most interesting, results in Table 4 suggest that participants' beliefs about other peoples' distancing behaviors predict their own willingness to engage in distancing and stay-at-home behaviors. We estimated this specification even though these latter findings linking incentivized beliefs to willingness to distance should be interpreted with caution. Theoretically, distancing and stay-at-home behavior is affected by one's preferences—a portion of which is captured by conservatism and the individual difference measures, which are fixed (i.e., pre-determined). However, whether one engages in these mitigation behaviors may be driven too by one's beliefs, which are malleable, and appear to react to charisma. Because there may be unmodeled variation in beliefs that determines distancing too, there is potential endogeneity problem in a model regressing distancing on beliefs. As we showed above, beliefs are affected by charisma, which is exogenous (i.e., manipulated).

Given the endogeneity problem and the fact that willingness to engage in distancing and stay-at-home behavior is elicited in a non-costly manner, we are faced with a hard methodological issue to tackle (i.e., individuals may claim they would follow public health guidelines regardless, because it is socially desirable to report this position). For this reason, we will give most weight to their true *belief* of what they think others would do following exposure to the speech. Irrespective of how they report being affected by the treatment, we can ensure that we capture at least in part how charisma affects their beliefs about

Table 4
IV results from the experiment (first and second stage): Effect of belief on willingness to stay at home.

	(1) DV: distancing	(2a) DV: belief	(2b) DV: distancing	(3a) DV: belief	(3b) DV: distancing	(4) DV: distancing
Male	−.36*	.07	−.37*	.07	−.37*	−.31*
	(2.22)	(.76)	(2.35)	(.80)	(2.35)	(2.01)
Age	.02**	.01†	.02**	.01*	.02**	.03***
	(2.89)	(1.88)	(2.81)	(1.99)	(2.81)	(4.26)
CRT	.01	.02	.02	.02	.02	−.03
	(.14)	(.22)	(.19)	(.25)	(.19)	(.30)
Over rater		3.08***		3.08***		
		(11.60)		(12.24)		
CRT*Over rater		−.19†		−.19†		
		(1.71)		(1.83)		
Conservative	−.74***	.09	−.74***	.09	−.74***	−.67***
	(7.64)	(.73)	(7.94)	(.73)	(7.94)	(4.12)
Conservative*Over rater		.19†		.20†		
		(1.75)		(1.92)		
Non-charismatic		−.15		−.16		−.03
		(.82)		(.89)		(.18)
Non-charismatic*Conservative		−.29*		−.29**		−.13
		(2.59)		(2.73)		(.64)
Non-charismatic*Over rater		−.02		−.02		
		(.13)		(.11)		
Belief	**.30*****		**.34*****		**.34*****	
	(5.15)		(5.42)		(5.38)	
State fixed effects	Incl.	Incl.	Incl.	Incl.	Incl.	Incl.
R-squared	.23	.65	.23	.65	.23	.18
	Comparison of non-linear and linear combination of estimators[a]: Effect of non-charismatic for:					
Liberals					**.04**	**.09**
					(.63)	(.47)
Moderates					**−.05**	**−.03**
					(.87)	(.18)
Conservatives					**−.15***	**−.16**
					(2.03)	(.49)

Note: Heteroscedastic robust *t*-statistics in parentheses $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, $^{\dagger}p < 0.1$; $n = 661$; baseline treatment condition for all models is Cuomo charismatic; [a]for Model 3 it is the indirect effect via beliefs; for Model 4 it is the reduced form effects. The Wald test for the joint difference of the bolded coefficients between Models 3 and 4 is $\chi^2(3) = 1.71, p = .64$. Estimation by OLS (Model 1), 2SLS (Model 2), Maximum Likelihood instrumental variable (Model 3), Maximum Likelihood (Model 4). Note, including the variable "non-charismatic" in the second-stage equation does not alter estimates or significance levels. CRT = Cognitive Reflection Task.

what others do, which could feed into what they themselves are willing to do. We interpret marginal effects for charisma as a function of conservatism (political ideology), which is one of our primary interests.

To estimate the effect of beliefs about others' distancing and stay-at-home behavior, and given that beliefs is an endogenous variable in Table 4 and that we manipulated the exogenous cause, we use instrumental variable regression to aim for estimating the causal effect (ERIV—experimentally randomized instrumental variable procedure, see Sajons, 2020; also, Bastardoz et al., 2023). Whatever estimator we used, we found beliefs to be strongly related to participants' own willingness to engage in distancing and stay-at-home behavior. In Model 1, we show the OLS coefficient for the effect of belief on willingness to distance and stay at home (*coef.* = .30, *SE* = .06, $z = 5.15$, $p < .0001$). Then in Model 2 we used instrumental-variable regression (i.e., two-stage least squares, 2SLS): We first model belief (Model 2a: the "first-stage" model) as the endogenous predictor and in Model 2b (the "second-stage" model) we estimate the effect of the predicted—that is, the endogeneity purged value—of belief on the outcome. We re-estimate the same model using maximum likelihood estimation in Models 3a and 3b, which is a more efficient estimator and found similar results.

Note, we modeled as "excluded" instruments (i.e., instrumental variables that were only in the first stage equation) the following: (a) charisma treatment, given we presume the effect of charisma if funneled through beliefs—also, this instrument is exogenous given we manipulated it); (b) being an over rater, given that it was specifically elicited on the belief and which is predetermined (and thus functioning as a model-implied instrument, see Bollen, 2012), and as a consequence, (c) the interaction of being an over rater with the CRT score; (d) the interaction of being an over rater with the charisma treatment; and given that charisma was an excluded instrument (e) the interaction of the charisma score with conservatism. Note, an exogenous variable interacted with another variable logically results in an exogenous (random) variable; thus, the excluded instruments should be theoretically exogenous. The constraints we made by excluding these instruments from the second-stage equation resulted in 5 degrees of freedom, which we tested via an overidentification statistic. The result was not significant, indicating that the model constraints were tenable, and the model did not violate the exclusion restriction.

The results from Models 2a and 2b showed that beliefs have a strong effect on the willingness to distance and stay at home, (*coef.* = .34, *SE* = .06, $z = 5.42$, $p < .0001$); the overidentification test was insignificant, $\chi^2(5) = 3.51, p = .62$, suggesting that the exclusion restriction held (assuming the specification is correct). The instruments were very strong in the first stage, Sanderson-Windmeijer multivariate $F(6, 603) = 137.92$, $p < .0001$. We found very similar results when using maximum-likelihood estimation.

Note, the indirect effect of the interaction of being non-charismatic and conservatism on willingness to distance and stay at home via beliefs, that is, the nonlinear combination of estimators of $\delta_1(\beta_1 + \beta_2 * \text{conservative})$, where the value of conservative is either -1 (liberal), 0 (moderate), or 1 (conservative), shows that only individuals who identify as politically conservative are affected by the manip-

ulations via beliefs, *coef.* = −.15, *SE* = .07, *z* = 2.03, *p* = .04 (see bottom panel of Table 4).

To reiterate, the instrumental variable results of Table 4 need to be interpreted with caution, because the exclusion restriction rests on the assumption that the interaction of being non-charismatic and conservatism affects willingness to engage in distancing and stay-at-home behavior *only* through the beliefs, and that it does not have an effect via another channel. Whereas this identifying assumption may be hard to defend in the absence of controls, in our setting we control for a battery of individual characteristics. We assume that the time-invariant (and in the short-run arguably hard-wired) preferences and attitudes are filtered out, and that the remaining variation in participants' willingness to engage in distancing and stay-at-home behavior may to a substantial extent be affected by—in the short-run more malleable—beliefs.

Note too, there is no difference in the coefficients between the reduced form model (Model 4) and the instrumental variable model with respect to the interaction effect. However, the reduced form model is very noisy (i.e., inefficient, with large measurement error), which would be expected in our modeling procedure, given that the CRT and being an over rater are powerful predictors of beliefs and substantially reduce the error variance in the "beliefs equations"; however, these variables are not relevant to the entire population under study per se in the distancing equation. Moreover, the reduced form model estimates the intention to treat (ITT), whereas the instrumental-variable model estimates the local average treatment effect (LATE)—where the portion of the population (politically conservatives) that were affected by the treatment plays a key role (Angrist, Imbens, & Rubin, 1996). These LATE estimates are relevant in the context of our study because we cannot model the efficiency of this estimator in the reduced form.

Brief discussion

The results of the experimental study show that individuals who identify as politically conservative are more likely to believe that a charisma signal will influence others' proclivity to engage in distancing and stay-at-home behaviors, and that this belief appears to drive whether participants themselves are willing to distance and stay at home when asked to. Individuals who identify as politically liberal seem to be unaffected. Although distancing and stay-at-home behavior were not directly observed, these results are interesting and corroborate somewhat the finding from the field data with respect to politically conservative individuals. Of course, the laboratory setting is hypothetical but insofar as beliefs are concerned, this measure was elicited in a costly manner. Still the results are interesting per se, though more research is required to make definitive conclusions about how charisma can be harnessed to solve coordination problems with respect to public health crises.

General discussion and conclusion

Our results suggest that charismatic communication may be able to save a substantial number of human lives in the fight against a pandemic. Using population mobility data, the results from our field study indicated that higher US governor charisma scores in COVID-19 speeches were associated with an immediate reduction in geographical mobility, a strong predictor of community spread of SARS-CoV-2. Results from our lab experiment provided individual-level evidence on the same relation, suggesting that politically conservatives were particularly receptive to charismatic communication.

Our results align with experimental evidence on the effectiveness of charismatic communication from organizational studies (e.g., Antonakis et al., 2021; Meslec et al., 2020) to suggest that the stylistic expression of leader communication holds great potential to coordinate individual behavior and facilitate public goods or increase individual effort. Our results also extend existing empirical studies on charisma in several ways.

First, we assess the large-scale implications for geographical mobility at the state level. The effect of charisma is not confined to coordinating behaviors among actors who are members of well-defined entities (like an organization), but also serves as a means for coordinating the actions of entirely fragmented groups, like the population of a state, towards the facilitation of a public good—here, public health. Second, our results showcase the potency of charisma for political leadership, an area that traditionally is either dominated by questionnaire research, which is severely limited (Fischer & Sitkin, 2023; Fischer et al., 2020) or by single, historical, or narrative accounts of individual leaders, rather than rigorous empirical evidence geared towards causal identification.

Third and finally, we add nuance to theories on charismatic leadership by shedding light on the complexities of individual responsiveness to charisma signals, at least in the context of a public health crises. In our case, political ideology seems to represent a crucial factor for subjects' receptivity to charisma. However, the saliency of various individual-level characteristics for eliciting an emotional response to charismatic appeals likely depends on the empirical context, the message conveyed, and the messenger. An important quest for future research is therefore to explore the nuances of charismatic expression and their potential heterogeneous implications for individual and group behavior.

Limitations

Readers should be mindful of certain limitations in our findings. First, in Study 1, we measured charisma signaling and not the charismatic effect. However, given that higher charismatic signaling usually triggers the attribution of charisma, we think it reasonable to assume that the results we observed must have come via the charisma signals engendering the charismatic effect on citizens. Next, although Study 1 drew on a balanced panel of governor speeches, and hence was able to eliminate many potential confounders through governor and time fixed effects, we cannot definitively rule out potential endogeneity from unobserved, time-variant factors such as idiosyncratic state-specific events that coincided with governors' press briefings and tracking of state mobility patterns. Yet, such threats are probably unlikely given that these events needed to not only affect the geographical mobility within states (e.g., extreme weather), but also affect governors' charismatic delivery of COVID-19 briefings.

Second, we are limited in our ability to explain *how* charisma affects stay-at-home behavior at the state level. For instance, one could imagine that media coverage or attention to individual speeches would affect the transmission of charisma to collective behavior via individual motivation of belief regarding what they think others will do; the latter would suggest that charisma may foster a common identity and make salient how costly action can solve coordination problems in blunting COVID-19's effect (cf. Antonakis et al., 2021). Uncovering the mechanisms by which charisma affect group behavior provides for an interesting area of future research.

The fact that Study 1 analyses were at the state level represents another limitation in the sense that it leaves us unable to uncover the complexities and heterogeneity in individual responses to governor charisma. State-level effects of governor charisma on geographical mobility could be a function of a host of different mechanisms, none of which we were able to definitively disentangle in Study 1. However, in controlling for state share of Republican vote, proportion of political conservatives in the state, and the political party of the governor, we did see some patterns in this regard, that charisma appeared to have the strongest effect in Republican led states. However, politics and possible "reactance" (cf. Ma, Dixon, & Hmielowski, 2019) of citizens to positions that may contradict their beliefs might confound the results

(and this effect may be exacerbated by governors having a different ideology than some citizens). Thus, in Study 2, we measured key individual demographic characteristics such as individuals' political ideology without them knowing the political party affiliation of the governor's speech. Conducting both studies allowed us to examine the average impact of charisma on actual mobility (stay-at-home behavior), while exploring the importance of individual characteristics. To further strengthen the bridging of findings from Study 1 to Study 2, however, future research could manipulate not only governor charisma, but also governor party to further unpack the importance of sender-receiver political ideology congruence for the effect of charismatic appeals. Of note too is that charisma's direct effect on belief was not significant; only the interaction with conservatism was. We provided theoretical explanations as to this observation; however, there is a potential issue with respect to the instrumental-variable estimation procedure, which requires valid instruments: exogenous and strong. Although charisma fulfills the first requirement it fails on the second. Thus, we must assume that the rest of the instruments were valid given that identification relied on the strongest, though model implied instrument, being an "over-rater." Future research should attempt to uncover different and stronger instruments to retest our insights.

Finally, even though Study 1 used observed behavioral responses that can be objectively verified, the data may not be fully reliable given that some people do not own cell phones and that those who do may not carry their cell phones with them at times. Nonetheless, most, about 85%, of Americans do own a smartphone (Pew Research Center, 2021). Moreover, our data showed considerable day-to-day variation in stay-at-home behavior (see replication files), indicating that there was mobility in the cell phones. Future research could triangulate our results using different mobility measures, although the accuracy, scalability, and unobtrusive nature of smartphone-based movement seem hard to trump. That the data are on the dependent variable is less worrying, though, given that measurement errors, if random, do not affect consistency of estimation (Ree & Carretta, 2006). In contrast to Study 1, the observed behavioral response in the experiment was hypothetical, though we incentivized it (and hence it was consequential for participants). Willingness to distance and stay at home, and the belief about how likely other Americans are to engage in these behaviors are ultimately based on individuals' perceptions and preferences. Future research should try to adopt another kind of design, like a natural experiment design, where some exogenous shock could be used to estimate the causal effect on real individual behavior (Sieweke & Santoni, 2020). Whereas we arguably captured core dimensions of stay-at-home behavior, it can encompass many different types of behaviors. Future research may therefore also do well to assess the impact of charismatic communication for other dimensions of distancing and stay-at-home behaviors than the ones represented in Study 2.

Implications

Notwithstanding its limitations and subject to future replication, our study has several implications. Whereas most non-pharmacological interventions (NPI) to fight COVID-19 have a series of socio-economic costs (e.g., lockdowns threaten jobs, school closures may lead to human capital depletion and raising inequalities), stepping up the quality of leader communication comes at a significantly reduced cost. Of course, the cost of learning charisma is not zero; however, it is typically small when put in perspective with respect to potential lives saved, especially if those communicating are naturally charismatic. Our main results are even more salient in light of recent findings suggesting that "soft" requests (e.g., stay-at-home orders) might not be as effective as compelling businesses to shut (e.g., Brauner et al., 2020). Moreover, charisma is not a celestial gift, but can be learned (cf. Antonakis et al., 2011), and providing charisma training for governors, or other top-level politicians, thus represents an easy and cost-effective way to increase the appeal and compliance with public health messages.

As our individual-level experiment data showed, it is important to consider how the extent and the use of charisma in communication may best be varied by audience. In the context of the current pandemic, when speaking to an audience of progressive and well-informed constituents, who easily follow scientific recommendations, leader charisma may be irrelevant to them. In contrast, when faced with an audience of ideologically conservative citizens who may be skeptical towards elites and scientists in general, the tools of charismatic communication may be a great help to transmit public health recommendations. Our results are very promising in this sense as they indicate that charisma worked best among politically conservatives who may be among the most skeptical of mitigation behaviors, at least in the current epidemiological crisis. This news is encouraging to scientists and practitioners alike, because we expect charismatic communication not only to matter for promoting distancing and stay-at-home behavior, but equally for other fronts at which the COVID-19 pandemic is fought, such as the social acceptability of the approved COVID-19 vaccines (Blanchard-Rohner et al., 2021; Jensen, Ayers, & Koskan, 2022). This expectation is naturally subject to scientific inquiry and should be tested in future research to assess whether charisma, too, can help combat COVID-19 along other fronts such as increasing vaccine acceptability.

That said, we acknowledge that we examined the effects during the onset on the pandemic. The eventual scope and duration of the pandemic's toll on everyday life, the transmissibility and severity of the virus, the changing behavior of citizens, public health messages and policies, bound what effects we observed. Still, the fact that we detected a strong effect during a particular point in time measured in many different states having different policies, suggests that the effect of charisma is reasonably robust; indeed, history suggests that the effect of charisma traverses space and time.

To conclude, in times of threat and crisis—whether COVID-19 or beyond—leaders must step up and play their part. Leaders must reassure, guide actions, instill hope and belief, and coordinate the efforts of the many towards a common goal. Whether the enemy is visible, invisible, or thought to be invincible, history has shown that words matter. Although it might seem benign or perhaps even trivial to the untrained observer, the effects of charisma are evident in governors' ability to coordinate the actions of millions to safeguard and promote the public health of a nation. The challenge today, and for the future, is to get more citizens vaccinated agains a virus that is endemic, akin to the influenza virus, and we think that how messages by those in power are crafted matter. Obviously, words and speeches cannot solve all problems. But they usually are the key to coordinating individuals' actions. As famously noted by former U.S. President, Barack Obama:

> Don't tell me words don't matter. I have a dream; just words? We hold these truths to be self-evident, that all men are created equal; just words? We have nothing to fear but fear itself; just words? Just speeches? (Obama, Feb 16, 2008)

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.leaqua.2023.101702.

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